

Q
=

8. Consider the statement "If Oscar eats Chinese food, then he drinks milk."



- (a) Write the converse of the statement.
- (b) Write the contrapositive of the statement.
- (c) Is it possible for the contrapositive to be false? If it was, what would that tell you?
- (d) Suppose the original statement is true, and that Oscar drinks milk. Can you conclude anything (about his eating Chinese food)? Explain.
- (e) Suppose the original statement is true, and that Oscar does not drink milk. Can you conclude anything (about his eating Chinese food)? Explain.

Sol

given statement: $\alpha \rightarrow \beta$

st.

α : Oscar eats Chinese food.

β : Oscar drinks milk.

a) converse: $\beta \rightarrow \alpha$

$\hookrightarrow$ i.e) if Oscar drinks milk then Oscar eats Chinese food.

b) contrapositive: $\neg \beta \rightarrow \neg \alpha$

$\hookrightarrow$ i.e) if Oscar does not drink milk then he does not eat Chinese food.

c) contrapositive can only be false

$y \Leftrightarrow \text{only } y$

og statement is also false.

d) y og statement is true and oscar drinks milk, we can not conclude anything.

e) y og statement is true and oscar does not drink milk then oscar must not eat chinese food.

PTO
 $\longrightarrow$

